(i)

$$\begin{aligned} f(x) &= x^2 e^{-x^2} \\ f'(x) &= 2xe^{-x^2} - 2x^3 e^{-x^2} \\ &= 2xe^{-x^2}(1 - x^2) \end{aligned}$$

Since $e^{-x^2} > 0$

$$f'(x) = 0 \iff x(1 - x^2) = 0$$

$$\boxed{x = 0,\ \pm 1}$$

(ii)

$$\begin{aligned} \frac{d}{dx}\left(P(x)e^{-x^2}\right) &= P'(x)e^{-x^2} - 2xP(x)e^{-x^2} \\ &= e^{-x^2}\left(P'(x) - 2xP(x)\right) \end{aligned}$$

$$\begin{aligned} P'(x) - 2xP(x) &= x(x^2 - a^2)(x^2 - b^2) \\ &= x^5 - (a^2 + b^2)x^3 + a^2 b^2 x \end{aligned}$$

Assume $\quad P(x) = -\frac{1}{2}x^4 + px^2 + q$

$$\begin{aligned} P'(x) - 2xP(x) &= -2x^3 + 2px - 2x\left(-\frac{1}{2}x^4 + px^2 + q\right) \\ &= x^5 - (2p + 2)x^3 + (2p - 2q)x \end{aligned}$$

$$\begin{cases} 2p + 2 = a^2 + b^2 \\ 2p - 2q = a^2 b^2 \end{cases} \implies \begin{cases} p = \dfrac{a^2 + b^2}{2} - 1 \\ q = -1 + \dfrac{a^2 + b^2}{2} - \dfrac{a^2 b^2}{2} \end{cases}$$

$$\boxed{P(x) = -\frac{1}{2}x^4 + \left(\frac{a^2 + b^2}{2} - 1\right)x^2 - 1 + \frac{a^2 + b^2}{2} - \frac{a^2 b^2}{2}}$$

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